

# LEAP vs. MBIR-Torch

A practical comparison from limited A100 tests | 10 September 2026

**Main finding:** LEAP was substantially faster on the tested GPU workloads. MBIR-Torch showed better forward/transpose consistency in these tests. Neither result establishes universal superiority, large-scale scalability, or reconstruction quality.

## Measured GPU performance

| Volume / views / detector   | Forward: LEAP / MBIR | Forward + backward: LEAP / MBIR |
|-----------------------------|----------------------|---------------------------------|
| 48 x 48 x 48 / 48 / 96 x 96 | 1.37 / 6.65 ms       | 2.86 / 11.53 ms                 |
| 96 x 96 x 96 / 24 / 96 x 64 | 1.27 / 12.12 ms      | 2.74 / 28.42 ms                 |
| 32 x 32 x 32 / 96 / 96 x 96 | 1.29 / 6.85 ms       | 3.03 / 13.00 ms                 |

LEAP was approximately **4.9-9.5x faster for forward projection** and **4.0-10.4x faster for forward + autograd backward**. Measurements use synchronized, resident float32 inputs, 10 warmups and 30 repeats; construction is excluded. Values come from separate runs of the same cases. Native LEAP CUDA is compared with eager MBIR-Torch, with compilation and Triton disabled. These are not matched-accuracy or equally optimized comparisons.

## One distinct issue: the backprojector as a gradient

A differentiable projector needs its backward operation B to equal the transpose of its forward operation A:  **$\text{dot}(\mathbf{A}\mathbf{x}, \mathbf{y}) = \text{dot}(\mathbf{x}, \mathbf{B}\mathbf{y})$** .

**LEAP:** two small cone-beam configurations failed the adjoint test; nine larger case/seed checks failed finite differences despite passing the normalized adjoint metric. A standalone native-API reproducer matched the adapter outputs exactly. For one small case, explicit matrices gave  **$\|\mathbf{B} - \mathbf{A}^{\text{transpose}}\|_{\text{F}} / \|\mathbf{A}\|_{\text{F}} = 31.6\%$** . This is an operator discrepancy, not a reconstruction-error percentage. It conflicts with the supplied matched-transpose description for the tested configuration; the internal cause remains unresolved.

**MBIR-Torch:** adjoint checks passed in the tested cases. A targeted subtraction-free directional comparison passed 30/30 trials (worst relative error 6.89e-5; tolerance 1e-4). One float32 finite-difference check remained intermittently failing; diagnostics support cancellation amplified by small forward-output variability. This is not an unconditional pass of every test.

## What users can reasonably conclude

| Consideration               | Practical implication                                                                                                                                                                                |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GPU throughput              | LEAP is the faster measured implementation here. Benchmark the intended geometry and enabled optimizations before selecting either tool.                                                             |
| Gradient-based methods      | MBIR has stronger consistency evidence here. Verify LEAP's selected forward/backprojector pair before relying on it for exact gradients; this does not invalidate ordinary LEAP reconstruction use.  |
| Memory and large volumes    | No defensible winner yet. Torch allocator peaks omit LEAP native allocations; process snapshots are not peaks. Tests reached only 96-cubed volumes, with no large-volume or multi-GPU scaling study. |
| Accuracy and reconstruction | Adjoint consistency does not establish physical projection accuracy. No matched-quality reconstruction or time-to-quality comparison was performed.                                                  |

**Test environment:** one NVIDIA A100-SXM4-40GB; LEAP 1.26; MBIR-Torch 0.0.2; Python 3.12.14; Torch 2.13.0+cu132; CUDA 13.2; driver 595.71.05. MBIR used eager float32, Triton disabled, ROR disabled and TF32 disabled. Findings are version- and configuration-specific.

**Evidence:** supplied A100 smoke/repeatability logs (9-10 Sep 2026), LEAP installation recheck, native adjoint reproducer and explicit-matrix diagnosis. MBIR finite-difference diagnostic used three fresh processes. No independent GPU rerun was performed for this summary.